Lecture on Nash's embedding theorem

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The fundamental theorem of Riemannian geometry often refers to the existence and uniqueness of Levi-Civita connection. But it also refers to the result of John Nash [6] that *any Riemannian manifold is isometric to a smooth submanifold in a Euclidean space.* In particular, Riemannian manifold can be defined as a submanifold of Euclidean space with induced metric tensor.

We will prove Nash's theorem for compact manifolds (B1). This case is leading; the general case requires a sequence of additional elementary statements that is outlined in Exercise ???.

In the proof we use the so-called Günther trick — an essential simplification of the original proof that was found by Matthias Günther [2]. We also use several minor improvements found by Deane Yang, Terence Tao, and Ralph Howard [3, 8, 9].

A Induced metric

Choose a smooth n-dimensional manifold Ω; here and further, smooth means C^∞.

Recall that a field of bilinear symmetric forms, say g, on the tangent bundle $\mathrm{T}\,\Omega$ is called metric tensor. A metric tensor g is called Riemannian if it is positive; that is, if $g(\mathrm{X},\mathrm{X}) > 0$ for any tangent vector $\mathrm{X} \neq 0$.

We will consider $\mathbb{R}^q$ with standard Euclidean metric. Recall that derivative of $w\colon \Omega \to \mathbb{R}^q$ in direction of a tangent vector $\mathrm{X} \in \mathrm{T}_p\,\Omega$ is denoted by

$$\mathrm{X}w = dw(\mathrm{X}).$$

Let $w\colon \Omega \to \mathbb{R}^q$ be a smooth map. We say that g is induced metric tensor for w if

$$g(\mathrm{X},\mathrm{Y}) = \langle \mathrm{X}w, \mathrm{Y}w\rangle$$

for any two tangent vectors $\mathrm{X}, \mathrm{Y} \in \mathrm{T}_p\,\Omega$ at any point $p \in \Omega$.

More generally, suppose $v, w\colon \Omega \to \mathbb{R}^q$ are two smooth maps. Let us write $g = Q(v, w)$ if

$$g(\mathrm{X},\mathrm{Y}) \equiv \tfrac{1}{2}\cdot[\langle \mathrm{X}v, \mathrm{Y}w\rangle + \langle \mathrm{X}w, \mathrm{Y}v\rangle].$$

The operator Q is symmetric and bilinear; it takes two smooth maps $v, w\colon \Omega \to \mathbb{R}^q$ and spits a metric tensor $g = Q(v, w)$ on Ω.

Note that if v and w are C^k-smooth, then $Q(v, w)$ is C^{k-1}-smooth. Observe that *a metric tensor g is induced by a smooth map $w\colon \Omega \to \mathbb{R}^q$ if and only if $g = Q(w, w)$.*

The components of $g = Q(w, w)$ in a local chart can be written as

$$g_{ij} = Q_{ij}(w, w) = \langle \partial_i w, \partial_j w \rangle.$$

A1. Exercise. *Let w be a smooth map from a smooth Riemannian manifold (Ω, g) to the Euclidean space $\mathbb{R}^q$. Show that g is induced by w if and only if w is length-preserving; that is, if*

$$\operatorname{length}\gamma = \operatorname{length}(w \circ \gamma)$$

for any curve γ in Ω.

The map $w\colon (\Omega, g) \to \mathbb{R}^q$ as in the exercise are often called isometric, but this term might be for other types of maps.

Let Ω be a smooth manifold. Given two smooth maps $v_1 : \Omega \to \mathbb{R}^{q_1}$ and $v_2 \colon \Omega \to \mathbb{R}^{q_2}$, denote by $w = v_1 \oplus v_2$ the map

$$w\colon \Omega \to \mathbb{R}^{q_1+q_2} = \mathbb{R}^{q_1} \oplus \mathbb{R}^{q_2}$$

defined by $w\colon p \mapsto (v_1(p), v_2(p))$.

A2. Exercise. *Show that*

$$Q(v_1 \oplus v_2, v_1 \oplus v_2) = Q(v_1, v_1) + Q(v_2, v_2)$$

for any two smooth maps $v_1 : \Omega \to \mathbb{R}^{q_1}$ and $v_2 : \Omega \to \mathbb{R}^{q_2}$.

Recall that support of a map is the closure of the set of its arguments with nonzero values.

A3. Exercise. *Let $v_1, v_2 : \Omega \to \mathbb{R}^q$ be smooth maps defined on a smooth manifold Ω. Suppose v_1 and v_2 have disjoint support. Show that*

$$Q(v_1 + v_2, v_1 + v_2) = Q(v_1 \oplus v_2, v_1 \oplus v_2).$$

B Formulation

The following theorem is the main result of this lecture.

B1. Theorem. *Let (Ω, g) be a compact Riemannian manifold. Then, for sufficiently large integer q, there is a smooth immersion $w\colon \Omega \to \mathbb{R}^q$ such that*

$$g = Q(w, w).$$

B2. Exercise. *Construct a metric tensor g on the torus $\mathbb{T}^2 = \mathbb{S}^1 \times \mathbb{S}^1$ such that $g(\mathrm{X}, \mathrm{X}) \geqslant 0$ for any tangent vector X, but there is no smooth map $w\colon \mathbb{T}^2 \to \mathbb{R}^{10}$ with the induced metric g.*

B3. Exercise. *Apply B1 and the Whitney embedding theorem to show that in B1 one can assume that w is an embedding. In other words, any compact Riemannian manifold is isometric to a submanifold in a Euclidean space.*

C Free maps

As before, Ω is a smooth n-dimensional manifold.

Recall that a C^1 map $f\colon \Omega \to \mathbb{R}^q$ is called regular if its differential df has rank n at each point. In other words, for any point $p \in \Omega$ and some (and therefore any) local coordinates at p all n first-order partial derivatives $\partial_1 f, \dots, \partial_n f$ are linearly independent.

A C^2 map $f\colon \Omega \to \mathbb{R}^q$ is called free (also known as two-regular) if an analogous property holds for first *and second* partial derivatives; that is, for any point $p \in \Omega$ and some (and therefore any) local coordinates at p all $n + \frac{n\cdot(n+1)}{2}$ partial derivatives $\partial_i f$, $\partial_{ij} f$ for $i \leqslant j$ are linearly independent.

C1. Exercise. *Show that there is no free map $\mathbb{S}^2 \to \mathbb{R}^4$.*

C2. Exercise. *Let $f_1\colon \Omega \to \mathbb{R}^{q_1}$ and $f_2\colon \Omega \to \mathbb{R}^{q_2}$ be smooth maps defined on a smooth manifold Ω. Suppose f_1 is free; show that so is $f_1 \oplus f_2$.*

Let f be a smooth function on a manifold Ω. Denote by E_f the smooth map $\Omega \to \mathbb{R}^2$, defined by

$$E_f(x) = (\cos[f(x)], \sin[f(x)]).$$

In other words, E_f is the composition of two maps

$$\Omega \xrightarrow{f} \mathbb{R} \xrightarrow{\ell} \mathbb{S}^1 \subset \mathbb{R}^2,$$

where $\ell\colon \mathbb{R} \to \mathbb{S}^1$ is the natural covering maps $y \mapsto (\cos y, \sin y)$.

C3. Exercise. *Consider the (x, y)-plane $\mathbb{R}^2$. Show that the map*

$$E_x \oplus E_y \oplus E_{x+y}\colon \mathbb{R}^2 \to \mathbb{R}^6 = \mathbb{R}^2 \oplus \mathbb{R}^2 \oplus \mathbb{R}^2$$

is free.

Generalize the statement to maps $\mathbb{R}^n \to \mathbb{R}^{n\cdot(n+1)}$.

C4. Exercise. *Let $f\colon \Omega \to \mathbb{R}^s$ be a regular smooth map and $F\colon \mathbb{R}^s \to$ $\to \mathbb{R}^q$ is free. Show that the composition $F \circ f\colon \Omega \to \mathbb{R}^q$ is free.*

Apply the Whitney embedding theorem and C3 to conclude that any smooth manifold admits a free embedding into a Euclidean space.

D Key lemmas: formulations

In this section we recall the definition of Hölder norm for functions on torus and formulate the Nash's and Günther's lemma.

Let us denote by $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ the n-dimensional torus. The torus comes with a global system of periodic coordinates which can be used to define the partial derivatives $\partial_1, \dots, \partial_n$. A metric tensor on $\mathbb{T}^n$ can be also written in these coordinates; so we can (and will) think that metric tensor is smooth map $g\colon \mathbb{T}^n \to \mathbb{R}^N$; here and further

$$N := \frac{n\cdot(n+1)}{2}.$$

Choose $0 < \alpha < 1$, say $\alpha = \frac{1}{2}$. Denote by $|x-y|_{\mathbb{T}^n}$ the standard distance between points $x, y \in \mathbb{T}^n$. Let us recall the definition of Hölder norms of a function $u\colon \mathbb{T}^n \to \mathbb{R}$

$$\begin{aligned} \|u\|_{0,\alpha} &:= \sup_{x\in\mathbb{T}^n}\{\,|u(x)|\,\} + \sup_{x,y\in\mathbb{T}^n}\left\{\frac{|u(x)-u(y)|}{|x-y|^{\alpha}_{\mathbb{T}^n}}\right\},\\ \|u\|_{k,\alpha} &:= \max_{|\mathcal{I}|\leqslant k}\{\,\|\partial_{\mathcal{I}}u\|_{0,\alpha}\,\}, \end{aligned}$$

where $\mathcal{I} = (i_1, \dots, i_n)$ denotes multi-index; so, $\partial_{\mathcal{I}} := \partial_1^{i_1}\dots\partial_n^{i_n}$ and $|\mathcal{I}| := i_1 + \dots + i_n$. The space of functions with finite norm $\|\ \|_{k,\alpha}$ is called the (k,α)-Hölder space; it is denoted by $C^{k,\alpha}(\mathbb{T}^n,\mathbb{R})$.

D1. Exercise. *Let $v \in C^\infty(\mathbb{T}^n,\mathbb{R})$. Show that there is a constant c such that*

$$\|u\cdot v\|_{k,\alpha} \leqslant c\cdot\|u\|_{k,\alpha}$$

for any $u \in C^{k,\alpha}(\mathbb{T}^n,\mathbb{R})$.

Further, let us define $C^{k,\alpha}(\mathbb{T}^n,\mathbb{R}^q)$ as the space of functions $u\colon \mathbb{T}^n \to \to \mathbb{R}^q$ with all coordinate functions from $C^{k,\alpha}(\mathbb{T}^n,\mathbb{R})$. Let

$$\|u\|_{k,\alpha} := \max_i\{\,\|u_1\|_{k,\alpha}, \dots, \|u_q\|_{k,\alpha}\,\},$$

where $u_1, \dots, u_q$ are the coordinate functions of u.

In the following two lemmas we fix a smooth free map $f\colon \mathbb{T}^n \to \mathbb{R}^q$ and let

$$L(w) := 2\cdot Q(f,w).$$

D2. Nash's lemma. *The linear operator*

$$L\colon C^\infty(\mathbb{T}^n,\mathbb{R}^q) \to C^\infty(\mathbb{T}^n,\mathbb{R}^N)$$

has a right inverse

$$M\colon C^\infty(\mathbb{T}^n,\mathbb{R}^N)\to C^\infty(\mathbb{T}^n,\mathbb{R}^q)$$

such that

$$\|Mh\|_{2,\alpha}\leqslant a\cdot\|h\|_{2,\alpha}$$

for some constant a.

D3. Günther's lemma. *There is a symmetric bilinear opearator*

$$\tilde{Q}\colon C^\infty(\mathbb{T}^n,\mathbb{R}^q)\times C^\infty(\mathbb{T}^n,\mathbb{R}^q)\to C^\infty(\mathbb{T}^n,\mathbb{R}^q)$$

such that

$$\begin{aligned}L\tilde{Q}&=Q,\\ \|\tilde{Q}(w,w)\|_{2,\alpha}&\leqslant b_2\cdot\|w\|^2_{2,\alpha},\\ \|\tilde{Q}(w,w)\|_{k,\alpha}&\leqslant b_2\cdot\|w\|_{2,\alpha}\cdot\|w\|_{k,\alpha}+b_k\cdot\|w\|^2_{k-1,\alpha}\end{aligned}$$

for any integer $k\geqslant 3$ and some fixed constants $b_2,b_3,\dots$

E Perturbation

E1. Perturbation lemma. *Let $f\colon\mathbb{T}^n\to\mathbb{R}^q$ be a smooth free map. Suppose $L(w):=2\cdot Q(f,w)$. Then there is $r>0$ such that the equation*

$$L(w)+Q(w,w)=h \tag{❶}$$

has a smooth solution $w\colon\mathbb{T}^n\to\mathbb{R}^q$ for any smooth metric h with $\|h\|_{2,\alpha}<r$. Moreover, we can assume that

$$\|w\|_{2,\alpha}\leqslant C\cdot\|h\|_{2,\alpha},$$

for some constant C.

Suppose that a metric tensor g is induced by a free map $f\colon\mathbb{T}^n\to$ $\to\mathbb{R}^q$. Note that the equation

$$Q(f+w,f+w)=g+h$$

is equivalent to ❶. Indeed,

$$\begin{aligned}Q(f+w,f+w)&=Q(f,f)+2\cdot Q(f,w)+Q(w,w)\\ &=g+L(w)+Q(w,w).\end{aligned}$$

Therefore the perturbation lemma implies the following.

E2. Corollary. *Let g_0 be induced by a smooth free map $f_0\colon \mathbb{T}^n \to \to \mathbb{R}^q$ and h another metric tensor on $\mathbb{T}^n$. If $\|h\|_{2,\alpha}$ is sufficiently small, then $g_1 = g_0 + h$ is induced by a smooth map $f_1\colon \mathbb{T}^n \to \mathbb{R}^q$.*

Proof of E1. Recall that M, a, $\tilde{Q}$, and b_2 are provided by D2 and D3. Set $R = \frac{1}{10\cdot b_2}$ and $r = \frac{R}{10\cdot a}$. Assume $\|h\|_{2,\alpha} \leqslant r$. By D3,

$$\Phi\colon w \mapsto Mh - \tilde{Q}(w,w)$$

is a contraction map from the closed ball $\overline{\mathrm{B}}[0,R] \subset C^{2,\alpha}(\mathbb{T}^n,\mathbb{R}^q)$ to itself.

Consider the sequence of maps $w_0, w_1, \dots\colon \mathbb{T}^n \to \mathbb{R}^q$ such that $w_0 = 0$ and

$$w_{n+1} = \Phi(w_n).$$

Note that w_n is smooth for any n. Since Φ is a contraction, w_n converges in $C^{2,\alpha}(\mathbb{T}^n,\mathbb{R}^q)$ as $n \to \infty$, say $w_n \to w$. Further,

$$\|w\|_{2,\alpha} \leqslant R \qquad \text{and} \qquad w = Mh - \tilde{Q}(w,w).$$

Since $LM = \mathrm{id}$ and $L\tilde{Q} = Q$, applying L to both sides of the last equation, we get that w solves ❶.

It remains to show that w is smooth. Suppose that

❷ *for an integer k, the sequence $\|w_n\|_{k,\alpha}$ is bounded.*

Since

$$\|w\|_{k,\alpha} \leqslant \varliminf_{n\to\infty} \|w_n\|_{k,\alpha}$$

it follows that $w \in C^{k,\alpha}$. Therefore it remains to prove ❷ for each integer $k \geqslant 2$. The base case $k = 2$ is already done; let us do the induction step.

Suppose $\|w_n\|_{k-1,\alpha}$ is bounded; in particular, there is a constant c_k such that

$$\|Mh\|_{k-1,\alpha} + b_k\cdot\|w_n\|^2_{k-1,\alpha} \leqslant c_k$$

for any n; here b_k is from D3.

By the second estimate in D3, we get

$$\begin{aligned}\|w_{n+1}\|_{k,\alpha} &\leqslant \|Mh\| + b_2\cdot\|w_n\|_{2,\alpha}\cdot\|w_n\|_{k,\alpha} + b_k\cdot\|w_n\|^2_{k-1,\alpha} \leqslant \\ &\leqslant c_k + b_2\cdot R\cdot\|w_n\|_{k,\alpha} \leqslant \\ &\leqslant c_k + \tfrac{1}{10}\cdot\|w_n\|_{k,\alpha}.\end{aligned}$$

In particular,

$$\|w_n\|_{k,\alpha} \leqslant 2\cdot c_k \qquad \Longrightarrow \qquad \|w_{n+1}\|_{k,\alpha} \leqslant 2\cdot c_k$$

for any n. Since $w_0 = 0$, induction on n implies ❷ for k. □

F Proof of Nash's lemma

Proof of D2. Recall that we have global periodic coordinates on $\mathbb{T}^n$.

Choose a metric tensor h on Ω. Since all $\partial_i f$, $\partial_{ij} f$ are linearly independent, the following identities define uniquely a vector field w on $\mathrm{T}^2\,\Omega$ that is spanned by $\partial_i f$, $\partial_{ij} f$

$$\langle \partial_i f, w\rangle = 0, \qquad\qquad \langle \partial_{ij} f, w\rangle = h_{ij}$$

for all i and j.

Since

$$\begin{aligned} L_{ij}(w) &= 2{\cdot}Q_{ij}(f,w) = \\ &= \langle \partial_i f, \partial_j w\rangle + \langle \partial_j f, \partial_i w\rangle = \\ &= \partial_j\langle \partial_i f, w\rangle + \partial_i\langle \partial_j f, w\rangle - 2{\cdot}\langle \partial_{ij} f, w\rangle, \end{aligned}$$

it defines the needed right inverse $M\colon h \mapsto w$. □

Let $f\colon \Omega \to \mathbb{R}^q$ be a regular smooth map and $p \in \Omega$. Denote by T_p the n-dimensional space spanned by the first-order partial derivatives $\partial_1 f, \dots, \partial_n f$ at p. Denote by $\mathrm{T}_p^\perp$ the normal complement of T_p in $\mathbb{R}^q$. We will denote by $\mathrm{T}\,\Omega$ and $\mathrm{T}_p^\perp\,\Omega$ the corresponding bundles over Ω.

Further, denote by T_p^2 the space spanned by all first and second-order partial derivatives $\partial_i f$, $\partial_{ij} f$ at p. Let $\mathrm{N}_p = \mathrm{T}_p^2 \cap \mathrm{T}_p^\perp$.

Note that that if f is free, then T_p^2 and N_p are respectively $(n + {} + \frac{n\cdot(n+1)}{2})$ and $\frac{n\cdot(n+1)}{2}$-dimensional at any point p. In this case we will denote by $\mathrm{T}^2\,\Omega$ and $\mathrm{N}\,\Omega$ the corresponding bundles over Ω.

Suppose $w = Mh$ as in the proof above. Since $\langle \partial_i f, w\rangle = 0$ for every i, we have $w(p) \in \mathrm{N}_p$ at any point $p \in \Omega$.

F1. Exercise. *Let $f\colon \Omega \to \mathbb{R}^q$ be a free map. Given a vector field v in $\mathrm{T}\,\Omega$ and a metric tensor h on Ω, show that there is a unique vector field w in $\mathrm{N}\,\Omega$ such that*

$$2{\cdot}Q(f, v + w) = h.$$

In other words, given v, there is a right inverse $M_v\colon h \mapsto v + w$ of L with w in $\mathrm{N}\,\Omega$.

G Proof of Günther's lemma

Note that many symmetric bilinear opearators $\tilde Q$ satisfy $L\tilde Q = Q$. In particular, one can take $\tilde Q = MQ$, where M is provided by Nash's lemma (D2). However, this choice of $\tilde{Q}$ does not satisfy the inequalities in the Günther lemma. The actual construction of $\tilde{Q}$ uses the extra freedom in choosing M that comes from F1.

Recall that we use global system of periodic coordinates on $\mathbb{T}^n$ to define the partial derivatives $\partial_1, \dots, \partial_n$. In particular, we can define the Laplacian

$$\Delta u = \sum_s \partial_{ss} u.$$

The next statement easily follows from the so-called Shauder estimates — a result in elliptic partial differential equations. It could be also obtained directly using Fourier analysis on torus; we omit its proof.

G1. Proposition. *Let $k \geqslant 2$ be an integer and $0 < \alpha < 1$. Then*

$$(\Delta - 1)\colon C^{k,\alpha}(\mathbb{T}^n) \to C^{k-2,\alpha}(\mathbb{T}^n).$$

is a bi-Lipschitz isomorphism. In particular, the linear operator

$$(\Delta - 1)\colon u \mapsto \Delta u - u$$

has a Lipschitz inverse

❶ $$(\Delta - 1)^{-1}\colon C^{k-2,\alpha}(\mathbb{T}^n) \to C^{k,\alpha}(\mathbb{T}^n)$$

By ❶, $(\Delta - 1)^{-1}$ is a smoothing operator it swallows $C^{k-2,\alpha}$ and spits $C^{k,\alpha}$. Note that $(\Delta - 1)^{-1}$ commutes with partial derivatives; that is,

$$(\Delta - 1)^{-1}\partial_i = \partial_i(\Delta - 1)^{-1}.$$

These two properties of $(\Delta - 1)^{-1}$ will be important in the following proof.

Proof of D3. Recall that $Q_{ij}(w,w) = \langle \partial_i w, \partial_j w\rangle$. Let us show that

❷ $$(1 - \Delta)Q_{ij}(w,w) = A_{ij}(w) + \partial_i A_j(w) + \partial_j A_i(w),$$

where each A_{ij}, A_i and A_j is a linear combination of the following quadratic forms

❸ $$w \mapsto \langle \partial_a w, \partial_b w\rangle, \quad w \mapsto \langle \partial_{ab} w, \partial_c w\rangle, \quad w \mapsto \langle \partial_{ab} w, \partial_{cd} w\rangle.$$

for all possible indexes a, b, c, d.

Indeed,

$$\begin{aligned}(1-\Delta)Q_{ij}(w,w) = \langle\partial_i w,\partial_j w\rangle-&\\ -\sum_s\big[2\cdot\langle\partial_{is}w,\partial_{js}w\rangle+\langle\partial_{iss}w,\partial_j w\rangle+\langle\partial_i w,\partial_{jss}w\rangle\big] =&\\ =\langle\partial_i w,\partial_j w\rangle-2\cdot\sum_s\langle\partial_{is}w,\partial_{js}w\rangle+2\cdot\langle\Delta w,\partial_{ij}w\rangle-&\\ -\partial_i\langle\Delta w,\partial_j w\rangle-\partial_j\langle\Delta w,\partial_i w\rangle.&\end{aligned}$$

Whence ❷ follows.

Choose $A=A_i$ or $A=A_{ij}$ for some i and j. Note that

❹
$$\begin{aligned}\|A(w)\|_{0,\alpha}&\leqslant c_2\cdot\|w\|^2_{2,\alpha},\\ \|A(w)\|_{k-2,\alpha}&\leqslant c_2\cdot\|w\|_{k,\alpha}\cdot\|w\|^2_{2,\alpha}+c_k\cdot\|w\|^2_{k-1,\alpha}.\end{aligned}$$

for some fixed constants $c_2,c_3,\dots$ Indeed, it is sufficient to check inequalities ❹ for each form in ❸, and the latter is straightforward.

Applying ❶, we get

❺
$$\begin{aligned}\|(1-\Delta)^{-1}A(w)\|_{2,\alpha}&\leqslant c_2\cdot\|w\|^2_{2,\alpha},\\ \|(1-\Delta)^{-1}A(w)\|_{k,\alpha}&\leqslant c_2\cdot\|w\|_{k,\alpha}\cdot\|w\|_{2,\alpha}+c_k\cdot\|w\|^2_{k-1,\alpha};\end{aligned}$$

the constants $c_2,c_3,\dots$ might be different from those above, but each of them depends only on f.

Further, recall that

❻
$$L_{ij}(w)=-2\cdot\langle\partial_{ij}f,w\rangle+\partial_i\langle\partial_j f,w\rangle+\partial_j\langle\partial_i f,w\rangle;$$

see Section F. Let us define $\tilde Q$ as a linear combination of $\partial_i f$ and $\partial_{ij}f$ for all $i\leqslant j$ such that

❼
$$\begin{aligned}-2\cdot\langle\partial_{ij}f,\tilde Q(w,w)\rangle&=(\Delta-1)^{-1}A_{ij}(w),\\ \langle\partial_i f,\tilde Q(w,w)\rangle&=(\Delta-1)^{-1}A_i(w)\end{aligned}$$

for all i and j. Since f is free, $\tilde Q$ is well defined. Combining ❻, ❼, and ❷, we get

$$L\tilde Q=Q.$$

Denote by e_i and e_{ij} the frame dual to $\partial_i f$ and $\partial_{ij}f$ for all $i\leqslant j$ in the space spanned by these partial derivatives. Note that

$$\tilde Q(w,w)=\sum_i(\Delta-1)^{-1}A_i(w)\cdot e_i-\tfrac12\cdot\sum_{i\leqslant j}A_{ij}(w)\cdot e_{ij}.$$

Since the maps $e_i,e_{ij}\colon\mathbb{T}^n\to\mathbb{R}^q$ are smooth, ❺ and D1 imply the last inequality in the lemma for each integer $k\geqslant 3$. □

H Nash's twist

In this section we prove the following statement which can be regarded as an approximate version of our main theorem (B1).

H1. Proposition. *Let g be a Riemannian metric on $\mathbb{T}^n$. Then there is a metric tensor h on $\mathbb{T}^n$ and one-parameter family of smooth maps $w_t\colon \mathbb{T}^n \to \mathbb{R}^q$ for $t > 0$ such that*

$$Q(w_t, w_t) = g + t{\cdot}h$$

for any t.

It might be tempting to pass to the limit of w_t as $t \to 0$, but, as you will see, this limit is constant; so it does not solve B1. The main player in the proof of the proposition is the so-called Nash's twist which are about to define.

Let φ and f be smooth functions on a smooth manifold Ω. Given $r > 0$, denote by $\mathbb{S}^1_r$ the circle of radius r in $\mathbb{R}^2$. The Nash's twist $F\colon \Omega \to \mathbb{R}^2$ of the triple (r, φ, f) is the following composition

$$\Omega \xrightarrow{f} \mathbb{R} \xrightarrow{\ell_r} \mathbb{S}^1_r \xrightarrow{\times\varphi} \mathbb{R}^2,$$

where $\ell_r\colon \mathbb{R} \to \mathbb{S}^1_r$ is a length-preserving covering map, say

$$\ell_r(x) = (r\cdot \cos \tfrac{x}{r}, r\cdot \sin \tfrac{x}{r}),$$

and $\times\varphi$ denotes the multiplication by φ; so

$$F(x) := \varphi(x){\cdot}(\ell_r \circ f(x)).$$

Recall that $(df)^2$ denotes the metric tensor induced by defined by $f\colon \Omega \to \mathbb{R}$; equivalently,

$$(df)^2(\mathrm{X}, \mathrm{Y}) := df(\mathrm{X}){\cdot}df(\mathrm{Y}) = (\mathrm{X}f){\cdot}(\mathrm{Y}f).$$

H2. Claim. *The metric tensor*

$$\varphi^2{\cdot}(df)^2 + r^2{\cdot}(d\varphi)^2$$

is induced by Nash's twist F for (r, φ, f).

Computations. Suppose that X is a tangent vector on Ω, then

$$\begin{aligned}\mathrm{X}F &= \mathrm{X}(\varphi{\cdot}\ell_r \circ f) = \\ &= (\mathrm{X}\varphi){\cdot}(\ell_r \circ f) + \varphi{\cdot}(\ell_r' \circ f){\cdot}(\mathrm{X}f).\end{aligned}$$

Observe that $|\ell_r| = r$, $|\ell'_r| = 1$, and $\ell_r \perp \ell'_r$. Therefore,

$$\langle \mathrm{X}F, \mathrm{Y}F\rangle = r^2\cdot(\mathrm{X}\varphi)\cdot(\mathrm{Y}\varphi) + \varphi^2\cdot(\mathrm{X}f)\cdot(\mathrm{Y}f). \qquad \square$$

Let $w\colon \Omega \to \mathbb{R}^q$ be a smooth map defined on a smooth manifold Ω. Suppose that $w_1, \dots, w_q$ are coordinate functions of w, then the induced metric tensor g can be written as

$$g = Q(v, w) = (dw_1)^2 + \dots + (dw_q)^2.$$

In order to prove B1, we need to find functions $w_1, \dots, w_q\colon \Omega \to \mathbb{R}$ that solves the above equation for a given Riemannian metric. The following proposition is a weaker form of B1; it will play a key role in the proof of H1.

H3. Proposition. *Let (Ω, g) be a compact n-dimensional Riemannian manifold. Then there are smooth functions $\varphi_1, \dots, \varphi_q, f_1, \dots, f_q$ on Ω such that*

❶ $$g = (\varphi_1)^2\cdot(df_1)^2 + \dots + (\varphi_q)^2\cdot(df_q)^2.$$

Proof. The metric tensor g can be written in local cart $(x_1, \dots, x_n)$ as

❷ $$g = \sum_{i,j} g_{ij}\cdot dx_i\cdot dx_j,$$

where $g_{ij} = g_{ji}$ are smooth functions of $(x_1, \dots, x_n)$.

Since g is Riemannian, we can choose chart so that ∂_i are orthonormal at a given point p; in other words, $g_{ii} = 1$ and $g_{ij} = 0$ at p for all $i \ne j$. Since g_{ij} are smooth, given $\varepsilon > 0$, we can choose a neighborhood $U \ni p$ such that

❸ $$g_{ii} \lessgtr 1 \pm \varepsilon \quad \text{and} \quad g_{ij} \lessgtr \pm\varepsilon$$

for all $i \ne j$ and any point in U.

Observe that

$$\pm dx_i\cdot dx_j = \tfrac{1}{2}\cdot(dx_i \pm dx_j)^2 - \tfrac{1}{2}\cdot(dx_i)^2 - \tfrac{1}{2}\cdot(dx_j)^2.$$

Let us plugin this formula in ❷ and take ❸ into account, assuming that ε is small. We get that g is a linear combination with positive coefficents of the metric tensors $(dx_i)^2$ and $(dx_i \pm dx_j)^2$ for all i and j. In other words, we can take the functions x_i and $x_i \pm x_j$ for all i and j as our functions $f_1, \dots, f_q$ and find $\varphi_1, \dots, \varphi_q$ such that ❶ holds in a neighborhood of p.

By the partition-of-unity, we get the global statement. □

In the presented proof, it is sufficient to take $2{\cdot}n^2$ functions in each chart; so, if Ω is covered by m charts, then one can take $q = 2{\cdot}m{\cdot}n^2$ functions. However, as it is stated in the following exercise, one may take q independent of m.

H4. Exercise. *Show that q in H3 can be bounded explicitly in terms of n; say, we can assume that $q = 100{\cdot}n^{100}$.*

Try to generalize H3 to noncompact manifolds.

Proof of H1. Suppose that $\varphi_1, \dots, \varphi_q, f_1, \dots, f_q$ are provided by H3. Let $r = \sqrt{t}$; denote by F_i the Nash's twist for the triple (r, φ_i, f_i). By H2,

$$w_t = F_1 \oplus \dots \oplus F_q$$

is the needed map with $h = (d\varphi_1)^2 + \dots + (d\varphi_q)^2$. □

H5. Advanced exercise. *Show that H1 holds for noncompact connected manifolds. Moreover, given any positive smooth function f on Ω, we can assume that h is bounded by $f{\cdot}g$; that is, $0 \leqslant h(\mathrm{x}, \mathrm{x}) \leqslant f{\cdot}g(\mathrm{x}, \mathrm{x})$ for any tangent vector x on Ω.*

I Proof assembling

I1. Exercise. *Let Ω be a submanifold in the torus $\mathbb{T}^n$. Show that for any Riemannian metric on Ω is a restriction of a Riemannian metric on $\mathbb{T}^n$.*

Proof of B1. By the Whitney embedding theorem, any smooth n-dimensional manifold Ω admits an immesion ι into $\mathbb{R}^{2\cdot n+1}$. Since Ω is compact, we can assume in addition that ι is short and its image lies in a small ball; if not, then we can pass to the rescaling of $\varepsilon{\cdot}\iota$ for small $\varepsilon > 0$. Composing ι with the covering map $\mathbb{R}^{2\cdot n+1} \to \mathbb{T}^{2\cdot n+1}$, we get an embedding $\Omega \hookrightarrow \mathbb{T}^{2\cdot n+1}$.

So from now on, we can assume that Ω is a compact submanifold of a torus.

By I1, any Riemannian metric g on Ω is the restriction of a Riemannian metric on the torus. Therefore,

❶ *it is sufficient to prove the theorem for $\Omega = \mathbb{T}^n$.*

Choose a free map $f\colon \mathbb{T}^n \to \mathbb{R}^q$; it exists by C3. Suppose g_1 is the metric tensor induced by f. We can assume that $g_1 < g$; that is, $g_1(\mathrm{x},\mathrm{x}) < g(\mathrm{x},\mathrm{x})$ for any tangent vector $\mathrm{x} \neq 0$. In particular, $g_2 = g - g_1$ is a Riemannian metric.

Applying J2, we get a metric tensor h on $\mathbb{T}^n$ such that for any $t > 0$ there is a smooth map $f_2\colon \mathbb{T}^n \to \mathbb{R}^{q_2}$ that induces $g_2 + t{\cdot}h$.

By the perturbation lemma, for any small $t > 0$ there is a smooth map $w\colon \mathbb{T}^n \to \mathbb{R}^{q_1}$ such that

$$Q(f_1 + w, f_1 + w) = g_1 - t{\cdot}h.$$

Finally by A2, the metric tensor

$$g = g_1 - t{\cdot}h + g_2 + t{\cdot}h$$

is induced by the map $(f_1 + w) \oplus f_2$. $\square$

I2. Exercise. *Let Ω be a connected noncompact manifold. Show that it admits a smooth proper bounded embedding in $\mathbb{R}^q \setminus \{0\}$. Moreover, given a Riemannian metric g on Ω, we can assume that the induced metric of this embedding is strictly smaller that g.*

J Remarks

Let us state another closely related result that shows a huge difference between C^1 and C^2 isometric embeddings. For example, it implies that the unit sphere admits C^1 length-preserving embedding into an arbitrarily small ball in Euclidean 3-space. There is no such C^2-embedding since Gauss curvature of the unit sphere is 1, but at an extremal point it must be at least $\frac{1}{r^2}$, where r is the radius of the ball.

J1. Nash–Kuiper theorem. *Let (Ω, g) be a n-dimensional Riemannian manifold and $f\colon (\Omega, g) \to \mathbb{R}^q$ be a short smooth regular map. Suppose that $q \geqslant n + 1$. Then for any $\varepsilon > 0$ there is an C^1-smooth length-preserving maps $f_\varepsilon\colon (\Omega, g) \to \mathbb{R}^q$ that is ε close to f; that is, $|f_\varepsilon(x) - f(x)| < \varepsilon$ for any $x \in \Omega$.*

Moreover if f is an embedding then we can assume that so is f_ε.

It was originally proved by John Nash [5] with the condition $q \geqslant n + 2$ and improved to $q \geqslant n + 1$ by Nicolaas Kuiper [4]. The original proof uses Nash's twist in a different way. Both papers are reader-friendly, it is better to start with the paper of Nash. One may also start with lectures by Allan Yashinski and the author [7] where related results were obtained using an alternative approach.

The discussed result formed a part of foundations of the so-called h o m o t o p y p r i n c i p l e, or h-p r i n c i p l e e; an excellent introduction is given in the book by Yakov Eliashberg and Nikolai Mishachev [1].

Many related questions are open. For example, it is unknown if *a neighborhood of any point in 2-dimensional Riemannian manifold admits a smooth length-preserving embedding into* $\mathbb{R}^3$.

This theorem is a leading partial case of Nash theorem on regular emebdding. The compactness condition can be removed; we will give a hint how to do it. Moreover the value q can be estimated in terms $n = \dim \Omega$. We will give a hint how to get a ruf bound, say $q = 100{\cdot}n^3$. The best known estimate so far is

$$q = \max\{ \tfrac{n\cdot(n+5)}{2}, \tfrac{n\cdot(n+3)}{2} + 5 \}.$$

In the following exercises we assume that $\varphi_1, \dots, \varphi_q, f_1, \dots, f_q$ are provided by H3 for a given Riemannian manifold (Ω, g). Further, F_i is the Nash's twist for the triple (r, φ_i, f_i), and

$$\begin{aligned} h &= (d\varphi_1)^2 + \dots + (d\varphi_q)^2, \\ F_r &= F_1 \oplus \dots \oplus F_q \colon \Omega \to \mathbb{R}^{2\cdot q}, \\ \Phi &= \varphi_1 \oplus \dots \oplus \varphi_q \colon \Omega \to \mathbb{R}^q. \end{aligned}$$

J2. Exercise. *Let (Ω, g) be a compact n-dimensional Riemannian manifold and $r > 0$.*

(a) Show that $g + r^2{\cdot}h$ is induced by F_r.

(b) Show that $F_r(p) \to 0$ as $r \to 0$ for any point $p \in \Omega$.

Let us denote by $\mathbb{R}^{r,s}$ the pseudoeuclidean space with signature (r, s); that is, the space $\mathbb{R}^{r+s}$ with scalar product defined by

$$\langle x, y\rangle = x_1{\cdot}y_1 + \dots + x_s{\cdot}y_s - x_{s+1}{\cdot}y_{s+1} - \dots - x_{s+r}{\cdot}y_{s+r},$$

where x_i and y_i denote the coordinates of vectors x and y in $\mathbb{R}^{r+s}$.

Show that g is induced by

$$F_r \oplus \Phi \colon \Omega \to \mathbb{R}^{2\cdot q, q} = \mathbb{R}^{2\cdot q, 0} \oplus \mathbb{R}^{0,q}.$$

The presented construction can be used to find an isometric immersion of Riemannian manifold (Ω, g) into a pseudoeuclidean space $\mathbb{R}^{r+s}$ with sufficiently large r and s.

J3. Exercise. *Let $f \colon (\Omega, g) \to \mathbb{S}^{q-1}$ be a smooth length-preserving embedding. Construct a smooth length-preserving embedding of any conformally equivalent manifold $(\Omega, \varphi^2{\cdot}g)$ into $\mathbb{R}^{q,1}$.*

That is, given a smooth positive function φ on Ω, construct a smooth map $\Omega \to \mathbb{R}^{q,1}$ with induced metric tensor $\varphi^2{\cdot}g$.

Semisolutions

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